13. If $\tan\alpha$ and $\tan\beta$ are the roots of $x^2+ax+b=0$, then $\dfrac{\sin(\alpha+\beta)}{\sin\alpha\sin\beta}$ is equal to

(1) $\dfrac{b}{a}$ (2) $\dfrac{a}{b}$ (3) $-\dfrac{a}{b}$ (4) $-\dfrac{b}{a}$

14. In a triangle ABC, $\sin^2 A+\sin^2 B+\sin^2 C=2$, then the triangle is

(1) equilateral triangle (2) isosceles triangle (3) right triangle (4) scalene triangle.

15. If $f(\theta)=|\sin\theta|+|\cos\theta|\,,\theta\in R$, then $f(\theta)$ is in the interval

(1) $[0,2]$ (2) $\left[1,\sqrt{2}\right]$ (3) $[1,2]$ (4) $[0,1]$

16. $\dfrac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$ is equal to

(1) $cos2x$ (2) $\cos x$ (3) $\cos 3x$ (4) $2\cos x$

17. The triangle of maximum area with constant perimeter $12m$

(1) is an equilateral triangle with side $4m$ (2) is an isosceles triangle with sides $2m,5m,5m$
(3) is a triangle with sides $3m,4m,5m$ (4) Does not exist.

18. A wheel is spinning at 2 radians/second. How many seconds will it take to make 10 complete rotations?

(1) 10π seconds (2) 20π seconds (3) 5π seconds (4) 15π seconds

19. If $\sin\alpha+\cos\alpha=b$, then $\sin 2\alpha$ is equal to

(1) $b^2-1,$ if $b\le\sqrt{2}$ (2) $b^2-1,$ if $b>\sqrt{2}$ (3) $b^2-1,$ if $b\ge 1$ (4) $b^2-1,$ if $b\ge\sqrt{2}$

20. In a $\triangle ABC$, if

(i) $\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}>0$

(ii) $\sin A\sin B\sin C>0$ then

(1) Both (i) and (ii) are true (2) Only (i) is true
(3) Only (ii) is true (4) Neither (i) nor (ii) is true.

Summary

Sum and Difference Identities(Ptolemy Identities):

$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta;\ \sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$

$\tan(\alpha+\beta)=\dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta};\quad \tan(\alpha-\beta)=\dfrac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}$